

ADYA KAR

Data Engineer

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PROFESSIONAL SUMMARY

- Experienced Data Engineer with 4+ years of expertise in building, maintaining, and optimizing scalable data pipelines using technologies like Python, Java, and SQL.
- Proficient in working with big data technologies, including Apache Kafka, Apache Spark, Hadoop (HDFS, MapReduce, Hive, Pig), and Apache Storm, to handle large-scale data processing and real-time analytics.
- Skilled in leveraging cloud platforms such as AWS (EC2, S3, Lambda, Glue, Athena, Redshift, CloudWatch) to implement data storage, processing, and warehousing solutions.
- Expertise in database management, with hands-on experience in MySQL, Oracle, MS SQL Server, and NoSQL databases like MongoDB and Pentaho, to design and maintain efficient data storage solutions.
- Strong background in ETL processes using tools like Apache Airflow, DBT, and Pentaho, ensuring efficient data extraction, transformation, and loading across multiple environments.
- Experienced in data warehousing and analytics using Amazon Redshift and Snowflake, enabling advanced business intelligence and reporting capabilities.
- Proficient in data analysis and modeling, including data cleaning, wrangling, clustering, forecasting, and ensuring data quality to derive actionable insights.
- Skilled in using data visualization tools like Tableau, Power BI, and MS Excel to create intuitive and insightful dashboards and reports.
- Familiarity with machine learning and deep learning libraries such as TensorFlow, scikit-learn, and PyTorch for implementing predictive analytics and advanced data models.
- Hands-on experience with DevOps tools like Kubernetes, Git, and the ELK Stack (Elasticsearch, Logstash, Kibana) for efficient data deployment, version control, and monitoring.
- Strong understanding of agile project management methodologies (Scrum, Kanban), with a proven track record of delivering data engineering projects in dynamic, fast-paced environments.

TECHNICAL SKILLS

Languages:	Python, Java, SQL
Libraries:	Pandas, NumPy, Apache Airflow, TensorFlow, scikit-learn, Matplotlib, Seaborn, Kafta, DBT, Hadoop, PySpark, Azure, Json, PyTorch
Databases:	MySQL, Oracle, MS SQL Server, NO SQL, MongoDB, Pentaho
Big Data Technologies:	Apache Kafka, Apache Storm, Apache Hive, Apache Spark, Hadoop (HDFS, MapReduce, Hive, Pig, Oozie)
Data Analysis:	Data cleaning, Data Modelling, Data warehousing, ETL, Clustering, Forecasting, Data Quality
Cloud Computing:	AWS (EC2, S3, Lambda, Glue, Athena, Redshift, CloudWatch)
Data Warehousing:	Amazon Redshift, Snowflake, Bigquery
Data Visualization Tools:	Tableau, Power BI, AWS Quicksight
Data Integration:	Restful API
Project Management:	Agile methodologies (Scrum, Jira, Kanban)
Other Tools:	Kubernetes, Git, Jira, ELK Stack (Elasticsearch, Logstash, Kibana)

PROFESSIONAL EXPERIENCE

AWS Data Engineer | CVS Pharmacy, USA

Sep 2023 – Current

- Develop and maintain **30+ ETL pipelines** using **Python, Java, and SQL** on **AWS Glue, AWS Lambda, and Step Functions**, processing **5TB+ of healthcare data daily**, ensuring **seamless integration across healthcare systems**.
- Utilized **Erwin** to create optimized **data models** and developed comprehensive **source-to-target mapping documents**, defining ingestion processes and transformations for structured and unstructured data.
- Implemented **real-time data ingestion** using **Spark Streaming, Kinesis and Lambda**, reducing **latency from 3 hours to 30 minutes** for **patient monitoring systems**.
- Built **automated data quality checks** in **Python & SQL**, reducing **data inconsistencies** and improving **reporting accuracy for clinical research**.
- Build and maintain data warehouses using **Amazon Redshift and Snowflake(dbt)** and **DynamoDB**, enabling healthcare organizations to perform advanced business intelligence, reporting, and predictive analytics.
- Developed **parameterized and reusable Jinja macros** in **DBT**, improving **modular ETL transformations** and reducing **code duplication**.
- Tuned **Snowflake warehouse configurations, query caching, and clustering strategies**, achieving **30% cost savings on compute resources**.
- Refactored **Apache Airflow DAGs** for **parallel execution and dynamic scheduling**, improving **pipeline run time by 40%** for **high-volume patient records processing**.
- Automated **ETL processes using Pentaho** and developed custom workflows with **python,java and Shell scripting**,contributing to more efficient data operations across the company.

- Conduct **data cleaning, data wrangling, and data quality assurance** to ensure the accuracy and reliability of healthcare data used for clinical decision-making and research purposes.
- Managed secure data access with IAM roles, ensuring compliance with security standards and controlling data access.
- Developed and published a **CDK package using Terraform** to automate the deployment of **VPC, VPC Peering, Load Balancer, Lambda functions**, reducing infrastructure provisioning time by 40% and streamlining DevOps workflows.
- Utilized tools like **Kubernetes, Git, and the ELK Stack (Elasticsearch, Logstash, Kibana)** for **containerization, version control, and monitoring** of data engineering solutions.
- Develop interactive dashboards and reports using **Tableau, Power BI, and AWS Quicksight**, providing healthcare stakeholders with visual insights and monitoring performance with Cloudwatch.
- **Designed and implemented a GenAI-driven intelligent data retrieval system** by integrating **OpenAI embeddings with a vector database (FAISS/Weaviate)** to enhance search

Data Engineer Everbridge India	Jul 2021 – July 2022
● Architected a scalable and high-performance data lake using AWS services (S3, Glue, Lake Formation, Athena, Redshift) for Everbridge’s incident management and notification systems. ● Designed and maintained Amazon Redshift & Snowflake-based data warehouses, optimizing query performance by 5x for medical research and patient outcome analysis. ● Automated workflows using Python, Java, Kornshell, and Jinja scripting, streamlining operations and reducing manual processes. ● Utilized Intellij and Pycharm IDEs for development and debugging, improving workflow efficiency and code quality. ● Designed scalable data storage solutions with MongoDB, MySQL, Oracle, and MS SQL Server, handling 100M+ patient records securely. ● Developed streaming analytics solutions using Spark & Kafka, enabling anomaly detection for high-risk patients with a 99.5% accuracy rate. ● Established automated data validation frameworks in diagnostic imaging and electronic health records (EHRs). ● Optimized AWS EMR & S3 storage, saving \$200,000 annually by transitioning to serverless architecture. ● Enforced HIPAA-compliant data processing with AWS IAM, VPC, and encryption standards, ensuring regulatory compliance. ● Designed and optimized relational and NoSQL databases, including MySQL, Oracle, MS SQL Server, and MongoDB, for managing patient records, medical imaging data, and other healthcare-related information. ● Integrate machine learning and deep learning models using TensorFlow, scikit-learn, and PyTorch to predict patient outcomes, identify high-risk patients, and optimize resource allocation.	

Data Platform Engineer Cognizant Technology Solutions India	Sep 2018 - May 2020
● Designed and implemented ETL architecture and Source-to-Target (S2T) mapping to load data into Data Marts supporting Comprehensive Capital Analysis and Review (CCAR) Reporting. ● Managed data ingestion from 23 Line of Business (LOB) servicing systems and 6 origination applications, integrating data into multiple Data Marts, including Servicing, Origination, Risk Exposure, and Credit Data Mart. ● Developed data pipelines using Informatica BDM & Sqoop, enabling seamless data ingestion from relational databases and mainframes into HDFS raw layer, followed by processing and transformation into Staging, UIR, and Mart Hive tables. ● Designed and optimized Spark-based ETL workflows, improving data processing efficiency and scalability. ● Developed Dimensional Data Models using Star and Snowflake Schemas to support analytical reporting and decision-making. ● Conducted data analysis and quality validation using Python (Pandas) and Informatica Data Quality (IDQ) to ensure accuracy and completeness of financial data. ● Provided technical guidance, mentoring, and leadership to ETL teams, facilitating effective collaboration in team and client meetings. ● Collaborated with cross-functional teams using Agile methodologies (Scrum, Kanban) to manage project timelines, deliverables, and ensure successful project outcomes.	

EDUCATION	
Texas A & M University	Aug. 2022 – Dec. 2023
Masters in Computer and Information Science, TX, USA	
College of Engineering and Technology	Aug. 2014 – May 2018
Bachelors in Information Science, Odisha, INDIA	

PROJECTS	
Fleet Safety Analytics Python, Hadoop, Power BI	March 2023
● Analyzed truck fleet safety metrics using Hadoop and Cloudera, focusing on geographical data, mileage, and fuel consumption, needed to enhance safety compliance and operational efficiency for truck fleets. ● Utilized Hadoop tools to process and analyze fleet data, integrated HDFS with Power BI for detailed descriptive analysis, Improved safety compliance by 20%.	
Uni Data Sync SQL, Linux, Cron	November 2022
● Developed a data processing pipeline to handle student enrollment data from multiple file systems (CSV, JSON, XML), ensuring seamless data integration and analysis. ● Scheduled data processing tasks using cron for regular updates and timely reporting, enhancing data availability and reliability.	